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DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring

A Internet of Everything Project
Presented to the Faculty of Information Technology
School of Communication, Multimedia & Computer Studies at
St. Dominic College of Asia City of
Bacoor, Cavite

In partial Fulfillment of Internet of Everything

Submitted By:

Khrystelle Jan Esplana

Misxa Bien Germino

Amira Rachel Marqueses

Gwyneth Ashley Mucio

Submitted to:

Dr. Alvin Jason Virata, MIT

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St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
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TABLE OF CONTENTS

Title Page ……1

Table of Contents……2

Chapter 1. Introductions

1.1Overview of IoT/IoE Project …… 4

1.2 Purpose/Objective of the Project …….. 5

1.3 Project Scope and Limitation…… 6

1.4 Significance and Impact of IoT/IoE Project …… 7

1.5 Key Terms and Definitions …….. 9

Chapter 2. Project Background and Motivation

2.1 Problem Statement ……12

2.2 Existing Solutions and Gaps …… 13

2.3 Project Motivation (Why your IoE project is the right solution?) ……15

Chapter 3. System Design and Architecture

3.1 System Overview……...17

3.2 Components Used (Devices, Sensors, etc)……19

3.3 Communication and Networking……...20

Chapter 4. Hardware and Software Implementation

4.1 Hardware Components (e.g., sensors, microcontrollers)……...22

4.2 Software/Platform Used ……23

4.3 System Integration (how the components work together)…… 24

Chapter 5. Project Development and Implementation Process

5.1 Project Planning and Design…… 26

5.2 Project Development Phase……27

5.3 Hardware Setup and Installation……... 29

5.4 Software Development and Configuration ……..30

5.5 System Testing and Validation……..31

5.6 Development and Integration……32

Chapter 6. Results and Findings

6.1 System Performance…….36

6.2 Challenges Faced and How They Were Solved……37

Chapter 7. Conclusion

7.1 Summary of the Project …….39

7.2 Future Work and Enhancement (Recommendations).....40

References……42

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CHAPTER 1

INTRODUCTION

1.1 Overview of IoT/IoE Project - Esplana, Khrystelle Jan G.

The DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring is designed to improve the process of monitoring attendance during seminars, events, and similar activities. Traditional attendance systems often rely on manual sign-in sheets, which can be time-consuming and may result in inaccurate records. Through the integration of IoT concepts and web-based technology, the proposed system provides a faster, more organized, and efficient method of managing attendance. The system connects RFID input devices to a web-based platform where attendance data is processed, stored, and accessed in real time.

The system primarily utilizes Radio Frequency Identification (RFID) technology to automatically record participant attendance. Each attendee is assigned an RFID-enabled identification card containing a unique identifier. Upon arrival at the event venue, participants scan their cards using a USB-based RFID reader connected to a computer or laptop. The RFID reader functions as a keyboard input device, allowing the system to automatically capture the scanned data and process it through the web application. The captured information, including participant identity and timestamp, is then sent to the backend server and stored in a cloud-based database.

Through the use of IoT connectivity and real-time data processing, the system allows event organizers to monitor attendance efficiently through an online dashboard. The recorded data can be viewed instantly, enabling administrators to track participant attendance, manage records, and analyze event participation. This approach reduces manual work, improves the

accuracy of attendance monitoring, and supports a more reliable and digital method of managing events.

1.2 Purpose/Objective of the Project - Germino, Misxa Bien D.

This study aims to design and develop DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring, intended to improve the efficiency, accuracy, and reliability of monitoring student and faculty participation during events at St. Dominic College of Asia.

To achieve this, the study aims to:

- Design and develop an RFID-based IoT attendance scanning node that allows students and faculty to log their attendance by tapping their RFID-enabled institutional identification cards.
- Implement a cloud-based data management system that securely stores and synchronizes attendance records transmitted from the RFID scanning hardware in real time.
- Develop a real-time attendance monitoring dashboard that enables administrators to view participant entry logs, monitor venue occupancy, and track participation levels during events.
- Create an attendance analytics module that archives historical attendance records and generates participation statistics for administrative review and event evaluation.
- Provide a centralized system for managing event attendance records to improve the accessibility and organization of attendance data for institutional reporting.

- Evaluate the developed system prototype in terms of functionality, reliability, efficiency, and usability in supporting event attendance monitoring within the Digital Campus of St. Dominic College of Asia.

1.3 Project Scope and Limitation - Marqueses, Amira Rachel

The goal of the DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring is to develop an automated attendance tracking solution that replaces manual, error-prone record-keeping with RFID technology. By integrating a USB-based RFID scanning device with a cloud-based backend, the system provides a seamless and efficient check-in experience for students and faculty while generating real-time attendance data and participation monitoring.

Scope:

- **RFID Attendance Scanning System**
  - A computer-based RFID scanning setup that validates student or faculty credentials through a USB 13.56 MHz Proximity ID Card Reader. When an RFID card is tapped, the system captures the unique identifier and records attendance automatically in the cloud database in real time.
- **Real-Time Attendance**
  - A centralized web-based interface that displays live attendance records, tracks participant activity, and stores historical data for administrative monitoring and evaluation.

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Limitation:

- **Dependence on Computer-Based Setup**
    - The system requires a computer or laptop to operate the USB RFID reader and run the web-based application. The absence of a dedicated device may limit portability compared to mobile or standalone systems.
- **Requirement for Physical Presence**
    - The system is designed for on-site attendance logging and it does not support remote or mobile-based check-ins. Participants must physically tap their registered RFID tags at the scanner node to log their presence.
- **Infrastructure Dependency**
    - The system relies on a stable internet connection to ensure real-time data synchronization with the cloud database. System performance may be affected during network interruptions or connectivity issues.
- **RFID Tag Frequency Compatibility**
    - The system is limited to RFID cards that are compatible with the USB 13.56 MHz Proximity ID Card Reader. Cards with different frequencies or unsupported formats may not be recognized by the system.

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1.4 Significance and Impact of IoT/IoE Project

DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring is a more reliable and technology-driven way to control event attendance at St. Dominic College of Asia - Digital Campus. Traditional attendance tracking methods frequently rely on human documentation, which can lead to incomplete data, administrative delays, and problems confirming participant participation. By combining RFID technology with cloud-based data processing, the system allows institutions to acquire attendance data more effectively while keeping organised digital records.

Additionally, the utilization of Internet of Things (IoT) technology makes it possible for digital and physical systems to cooperate in a networked setting. Through this connectivity, the RFID scanner's attendance data is instantly sent to the centralized database, where authorized administrators may view and keep an eye on it. In addition to increasing the accuracy of attendance records, this integrated procedure enables event planners to get precise participation data without the need for manual data gathering.

The project is expected to make a meaningful contribution to several groups inside the educational institution.

- **For Students and Faculty Members:** The method simplifies the attendance process at institutional events by allowing participants to authenticate their presence using RFID identity cards. This strategy reduces registration delays while also ensuring that attendance records are regularly captured. Participants also benefit from a better organised event management procedure, which allows their participation to be correctly validated.

- **For Event Organizers and Administrators:** The system offers a digital platform that helps administrators manage attendance activities. Using a centralized monitoring interface, organisers may more efficiently review attendance logs and manage event participation statistics. This eliminates the need for human record checking and allows organisers to concentrate on other critical parts of the event.
- **For Institutional Management:** Reliable attendance data can help with institutional reporting and event evaluation. By keeping structured digital records, the institution may analyse participation trends and decide which events garner the most attention. Such data could help guide planning decisions and improve the organization of future campus activities.
- **For Technological Development in Educational Systems:** This project shows how IoT technology can be used in academic settings to overcome operational difficulties. By combining RFID authentication devices, wireless connectivity, and cloud-based data management, the study provides a practical illustration of how linked technologies might help with smarter campus operations.

1.5 Key Terms and Definitions

To ensure a thorough understanding of the technical and operational components of this research, the following terms are defined:

Technical Terms:

- **RFID** (Radio Frequency Identification): A wireless technology that detects and retrieves profile data from institutional ID tags to enable safe user verification.

- **Web-Based System:** An application that is accessed through a web browser and utilizes internet connectivity to transmit, process, and store data between users and a centralized server.
- **Backend System**: The server-side environment that manages approval workflows, processes data, and maintains the administrative dashboard.
- **The IoT Framework**: An integrated system in which physical devices (RFID readers and ESP32s) communicate with cloud infrastructure to enable automated processes.
- **The Internet of Things (IoT):** A technological framework made up of smart devices that communicate and exchange data to improve functional workflows and the overall user experience.
- **The Internet of Everything (IoE):** A vast network that integrates human interaction, systemic processes, data flows, and physical objects to enable better informed decision-making and improved operational performance.

OPERATIONAL TERMS:

- **Event Attendance Monitoring:** Is the systematic process of recording participant presence at institutional events utilising an RFID-based scanning device.
- **RFID Attendance Scanner:** A hardware device that allows attendees to track their attendance by tapping their RFID-enabled identity cards during an event.
- **Attendance Monitoring Dashboard:** A digital interface that allows administrators to monitor attendance logs and view participation statistics generated by the RFID scanning equipment.

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- **Attendance Database:** A centralised storage system that securely stores all attendance records, including identification numbers and time logs.
- **Participant Identification Record:** A digital profile connected to an RFID card that enables the system to identify a participant and match their attendance data to the appropriate event record.
- **Participation Data Analysis:** A system-wide procedure for organising recorded attendance data so that administrators can compare participation levels across events.

CHAPTER 2

PROJECT BACKGROUND AND MOTIVATION

2.1 Problem Statement - Germino, Misxa Bien D.

Monitoring student and faculty attendance during institutional events is essential for ensuring accurate participation tracking and effective venue management. At St. Dominic College of Asia (SDCA), many academic and co-curricular activities are conducted across different venues within the Digital Campus, requiring organizers to monitor participant attendance efficiently. However, attendance monitoring for these events often relies on traditional methods such as manual sign-in sheets or basic digital forms. These approaches can result in inaccurate records, administrative delays, and difficulty in verifying the actual presence of participants. According to (Nkata 2024), traditional attendance recording methods are often inefficient and unreliable due to their susceptibility to human error, time-consuming processes, and challenges in verifying accurate student participation.

Event organizers also face difficulties in monitoring real-time participation and venue occupancy during large gatherings. Manual attendance systems make it difficult to immediately determine the number of participants present in a venue or retrieve attendance records for reporting and administrative review. Supporting this, (Salunkhe et al. 2025) emphasized that manual attendance management in educational institutions is time-consuming and prone to inaccuracies, highlighting the importance of automated technologies such as RFID-based attendance systems to improve efficiency and data accuracy. With recent advancements in Internet of Things (IoT) technologies and RFID-based authentication systems, attendance data can now be captured instantly and transmitted to cloud-based platforms for real-time monitoring and analysis.

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Despite these technological developments, many institutions still lack an integrated system capable of automatically recording event attendance and providing real-time participation analytics. Without such a system, organizers may experience difficulties in tracking attendance levels, managing venue occupancy, and maintaining reliable records of event participation. Therefore, this study proposes the development of DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring, which aims to provide a faster, more accurate, and real-time method of monitoring event participation and venue occupancy at St. Dominic College of Asia.

2.2 Existing Solutions and Gaps - Germino, Misxa Bien D.

In many educational institutions and organizations, event attendance monitoring is still conducted using traditional or semi-digital methods. These approaches often rely on manual attendance sheets, spreadsheets, or basic digital systems that lack automation and real-time tracking. As a result, issues such as inaccurate records, proxy attendance, delays in data processing, and limited visibility of participant attendance frequently occur. Although some digital solutions have been introduced, many systems still fail to provide efficient and reliable real-time monitoring.

Existing Solutions

Regardless of advancements in digital facility management, many institutions rely on systems that only partially address reservation and utilization concerns. The most common existing solutions include:

Manual Attendance Systems

- Some organizations still use physical attendance sheets or spreadsheet-based tracking.

- Attendance is recorded manually by facilitators.
- Monitoring and verification of attendance are done manually.

RFID-Based Attendance Systems

- RFID systems are commonly used for attendance tracking in events and institutions.
- Users tap identification cards to record their presence.
- Data is stored in a central database for record-keeping.

Web-Based Monitoring Systems

- Some institutions use web-based platforms to manage attendance data.
- Attendance information is displayed through dashboards.
- Data input is sometimes manual or not updated in real time.

Gaps in the Existing Solutions

Despite these technological developments, significant gaps remain in addressing the integration of reservation and actual room utilization monitoring. The identified gaps include:

Lack of Real-Time Monitoring

- Most systems do not provide immediate updates of attendance data.
- There is no continuous monitoring of participant presence during events.
- This results in delayed reporting and limited visibility.

Limited Integration of Technologies

- RFID systems are often not fully integrated with web-based platforms.
- Attendance data may not be synchronized across systems.
- This reduces efficiency and monitoring accuracy.

Inefficient Data Management

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- Manual and semi-digital systems are prone to human error and data inconsistencies.
- Duplicate or missing records may occur.
- Affects the reliability of attendance tracking.

How the DC Space Bridges These Gaps

DC Space addresses the limitations of existing systems by integrating RFID technology with a web-based IoT platform for real-time attendance monitoring. Unlike traditional systems that focus only on basic attendance recording, the proposed system connects RFID data capture with immediate processing and visualization.

By integrating RFID scanning with a centralized database, the system automatically records attendance and stores data securely. This reduces manual workload, minimizes errors, and ensures accurate record-keeping. The real-time monitoring feature allows administrators to track attendance instantly, improving transparency and efficiency. Through automated data processing and centralized management, the system provides a reliable, scalable, and user-friendly solution for event attendance monitoring.

2.3 Project Motivation (Why your IoT project is the right solution?)

When there are a lot of people attending institutional events, it might be challenging to control attendance. Many events continue to use antiquated techniques like manual sign-in sheets, which frequently need that organisers physically review and compile records following the event. These methods could result in handwriting that is difficult to see, missing entries, or delays in creating attendance records. Institutions need a more dependable method that can streamline attendance tracking while preserving accurate documentation as events continue to expand in size.

An effective way around these restrictions is to include RFID technology into an Internet of Things-based system. Attendance can be automatically recorded without the need for human input from event planners by enabling participants to verify their presence using RFID identification cards. The identifying data is scanned, then sent to the cloud system for processing and storage. This automated process guarantees that attendance data is recorded in an orderly and easily accessible way while lowering the possibility of recording errors.

Another feature of the suggested system is its capacity to provide real-time monitoring. Event organisers can check attendance data while the event is in progress, allowing them to monitor participation levels without waiting for handwritten reports. Furthermore, keeping digital records allows administrators to obtain prior attendance data as needed for reporting or evaluation purposes.

Overall, the DC Space IoT attendance monitoring system offers an organised, technologically advanced alternative to traditional attendance approaches. By merging RFID authentication, wireless connectivity, and cloud-based data administration, the project creates a more effective method of gathering and monitoring event attendance. This approach allows the institution to improve the reliability of its attendance monitoring practices while also encouraging the continued use of smart campus technologies.

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE

3.1 System Overview - Esplana, Khrystelle Jan G.

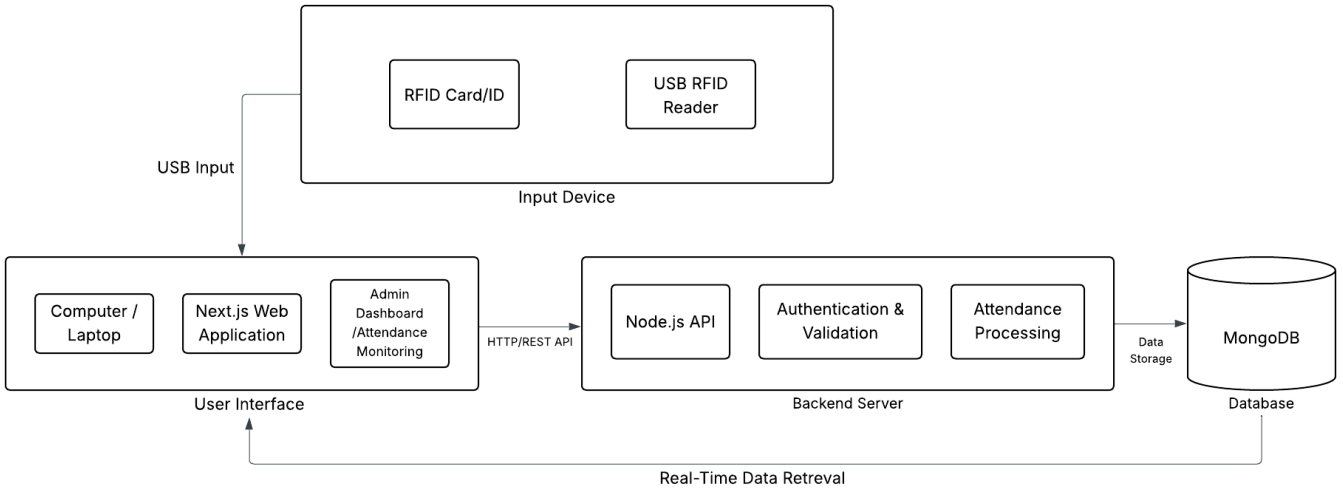


Figure I. Overall System Architecture of the DC Space IoT-Based Event Attendance Monitoring System

Figure I shows the overall system architecture of the proposed DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring. The system is composed of four main components: the input device, user interface, backend server, and database, which work together to automate the attendance monitoring process.

The process begins at the input device, where participants use their RFID cards or IDs to scan using a USB RFID reader. The RFID reader functions as a keyboard input device, which automatically captures the unique identifier of the card and sends it directly to the computer or laptop.

The captured data is then processed by the web-based application developed using Next.js. This serves as the user interface of the system, where administrators can access the dashboard for attendance monitoring and management. The web application receives the RFID input and prepares it for further processing.

The system then communicates with the backend server through HTTP or REST API requests. The backend, developed using Node.js, handles authentication and validation to ensure that the scanned RFID data corresponds to a registered participant. It also performs attendance recording by storing the time-in and time-out data.

After processing, the backend server stores the attendance records in the MongoDB database. The database serves as the central storage of all participant information and attendance logs. Finally, the stored data is retrieved from the database and displayed on the admin dashboard in real time. This allows administrators and event organizers to monitor attendance efficiently, view records instantly, and manage participant data during and after the event.

Overall, the integration of RFID technology with a web-based system and cloud database provides a faster, more accurate, and efficient solution for event attendance monitoring.

3.2 Components Used (Devices, Sensors, etc.) - Marqueses, Amira Rachel

A. Hardware Components

1. **USB 13.56 MHz Proximity ID Card Reader** – A plug-and-play RFID device used for scanning identification cards. It connects directly to a computer or laptop via USB and functions as a keyboard input device. When an RFID card is tapped, the reader automatically inputs the card's unique identifier (UID) into the system, enabling fast and accurate attendance recording without the need for a microcontroller.
2. **RFID Tags (Student ID)** - The passive tags embedded in the institutional ID cards. These contain the digital signatures required to authenticate the user and pull their specific profile data for the attendance of the event.
3. **Computer/Laptop Device** – Serves as the main processing unit of the system. It runs the web-based attendance application, receives input from the RFID reader, processes the scanned data, and communicates with the cloud database for real-time data storage and retrieval.
4. **Network Connectivity (Wi-Fi/Internet)** – Provides the necessary internet connection for transmitting attendance data from the local system to the cloud-based database. It ensures real-time synchronization and accessibility of attendance records through the admin dashboard.

B. Software Components

1. **RFID Input Processing Module** – Handles the input received from the USB RFID reader. Since the device functions as a keyboard, this module captures

the scanned UID from the input field and forwards it to the system for validation and processing.

2. **Authentication and Validation Module** – Responsible for verifying the scanned RFID UID against the stored database records. It ensures that only registered users are recorded and prevents duplicate or invalid entries.
3. **Scanner Interface & Sync Module** – Acts as the bridge between the RFID input and the cloud-based system. It processes incoming scan data and ensures that all attendance records are transmitted and updated in real time within the database.
4. **Real-Time Attendance & Analytics Engine** – Functions as the central nervous system of the project. It ingest real-time scan data to maintain a live, accurate count of attendees and venue occupancy. By processing these streams instantly, it provides an automated, error-free alternative to manual registration.

3.3 Communication and Networking - Marqueses, Amira Rachel

The effectiveness of the DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring relies on the seamless data exchange between the physical scanner hardware and the cloud backend. This connectivity ensures that attendance is captured in real-time, providing administrators and event organizers with immediate, live status updates. By utilizing wireless communication, the system creates a direct bridge between the RFID scanning node and the cloud infrastructure, enabling an automated, frictionless check-in experience for participants.

1. **Wireless Communication Technologies**

   To maintain a constant link between the scanner and the system's digital infrastructure, the project utilizes the following:

   - **Internet Connectivity (Wi-Fi) :** The computer running the web-based system connects to the internet through Wi-Fi. This connection enables the system to transmit and store attendance data in the web server database and allows real-time updates to be reflected in the admin dashboard.
   - **Web-Based System Access:** The attendance system is implemented as a web-based application that can run on a local server (localhost) or a hosted web server. This allows users and administrators to access the system through a web browser, making it flexible and accessible within the network.
   - **Remote Support Capabilities:** The system can be updated, patched, and checked for errors wirelessly, allowing it to stay fully functional without the need for manual, in-person troubleshooting.

These networking technologies create a secure, high-efficiency ecosystem for campus event management. The cloud-native architecture maintains continuous synchronization between the hardware and the analytics dashboard, delivering automated check-ins and real-time visibility into occupancy. Ultimately, this digitization empowers administrators with accurate metrics regarding event participation and venue usage.

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CHAPTER 4

HARDWARE AND SOFTWARE IMPLEMENTATION

4.1 Hardware Components - Misxa Bien Germino

To implement the physical identification process, the hardware modules were carefully installed and calibrated. Testing was conducted on the entire suite to ensure all components functioned correctly as a unified system during the attendance-tracking trials.

13.56 MHz USB RFID Reader. The 13.56 MHz USB RFID Reader was used as the primary input device for capturing attendance data in the system. This device reads the unique identification number embedded in RFID cards when tapped within close proximity. The reader was connected to the computer via a USB interface, enabling real-time transmission of attendance data to the system database. The device was configured to ensure accurate and fast detection of RFID tags during event registration and attendance monitoring.

RFID Cards / Student Identification Cards. RFID cards, specifically student identification (ID) cards, were used as unique identifiers for each registered participant in the system. Each student ID contains an embedded RFID chip that stores a unique identification number associated with the student's information in the database. When the student taps their ID card on the RFID reader, the system automatically captures the identification number and records the attendance in real time. This process improves accuracy, reduces manual recording errors, and enhances the efficiency of attendance monitoring during events.

Computers / Laptops. Computers or laptops were utilized to run the web-based attendance monitoring system and support the overall operation of the platform. These devices served as the primary interface for administrators and event organizers to manage attendance records, monitor participant data, and oversee system activities in real time. The

computers or laptops were connected to the RFID reader through a USB interface, enabling seamless communication between the hardware and the software components of the system. Additionally, these devices provided the necessary processing power and display interface required for efficient system performance and user interaction.

4.2 Software/Platform Used - Misxa Bien Germino

The software components of the system were implemented to enable seamless communication between the RFID input device, backend server, database, and user interface within the attendance monitoring system.

RFID Input Interface (13.56 MHz USB RFID Reader). The system utilizes a 13.56 MHz USB RFID Reader operating in keyboard (HID) mode. This allows the device to function as a standard input device that automatically transmits the unique identifier (UID) of an RFID card into the system upon scanning. This approach eliminates the need for additional drivers or specialized middleware, enabling straightforward integration with the frontend interface.

Node.js Backend Server. The backend server was implemented using Node.js to handle attendance recording and data retrieval. It was used to receive RFID UID inputs from the frontend, validate them against stored user records, and store corresponding attendance logs in the database. The backend manages the recording of timestamped entries such as Tap IN and Tap OUT, ensuring accurate tracking of user participation. It also handles requests from the frontend to retrieve and display attendance data, allowing users and administrators to view records in real time. All data processing is performed through API routes to ensure efficient and consistent communication between the frontend and the database.

Next.js Web Dashboard. The web-based dashboard was developed using Next.js and served as the main interface for real-time monitoring of attendance. It allows users to browse events, register, and view their attendance records dynamically. The dashboard captures RFID inputs through input fields and displays corresponding attendance logs instantly. It also provides administrators with visibility of attendance data as it is recorded, supporting real-time monitoring of event participation. The interface was designed to be responsive and user-friendly, ensuring efficient interaction and clear visualization of attendance status.

MongoDB Database. The database was used as the data storage system for managing user information, event records, and real-time attendance logs. It stored essential details such as RFID UID, student number, event schedules, and timestamped attendance entries. The database supports real-time monitoring by allowing immediate updates and retrieval of attendance data during RFID scanning. It was accessed through the backend server to ensure centralized and consistent data handling. Data updates, including attendance logs and event-related records, were observed to reflect accurately during system testing, ensuring reliability, scalability, and data integrity.

4.3 System Integration - Misxa Bien Germino

The system was integrated to establish a seamless connection between the RFID input device, web-based interface, backend processing, and database storage, forming a complete IoT-enabled attendance monitoring solution. Each component works together to ensure that user interactions are captured, processed, and reflected in real time within the system.

The process begins when a user scans their RFID card using the 13.56 MHz USB RFID reader. The device automatically inputs the card's unique identifier (UID) into the web

dashboard, where it is captured through a designated input field. The frontend then forwards this data to the backend server through API requests for verification.

Upon receiving the UID, the backend validates it against the stored user records in the database. If a match is found, the system records the corresponding attendance log with a timestamp, identifying whether it is a Tap IN or Tap OUT entry. The recorded data is immediately stored in the database, ensuring accurate and consistent record-keeping.

After processing, the backend sends a response back to the frontend, allowing the dashboard to display updated attendance information in real time. Users and administrators can then view attendance status and logs without delay. This integration highlights the system's ability to provide an efficient, automated, and real-time attendance monitoring process using RFID technology.

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CHAPTER 5

PROJECT DEVELOPMENT AND IMPLEMENTATION PROCESS

5.1 Project Planning and Design - Amira Rachel Marqueses

The development of the DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring began with careful planning and system conceptualization. The project aimed to improve traditional attendance monitoring by replacing manual sign-in methods with an automated and real-time digital solution. During the early stages, the team analyzed existing attendance processes used in institutional events and identified key issues such as time-consuming registration, human errors, and difficulty in tracking attendance records.

After identifying these problems, the team defined the system objectives, which include improving efficiency, ensuring accurate data recording, and enabling real-time attendance monitoring. The system was designed to be user-friendly for both participants and administrators while maintaining reliable data synchronization between hardware and the cloud-based system.

The design phase focused on selecting appropriate hardware and software components. Instead of using microcontrollers, the system utilizes a USB 13.56 MHz Proximity ID Card Reader as the primary input device. This reader connects directly to a computer or laptop and functions as a keyboard input device, allowing scanned RFID data to be automatically entered into the system.

On the software side, a web-based application was developed to manage attendance data. The system includes a cloud database for storing records and an admin dashboard for

monitoring attendance in real time. Internet connectivity plays a crucial role in ensuring that scanned data is transmitted and updated instantly.

System architecture diagrams and data flow models were created to visualize how data moves from RFID scanning to database storage and dashboard display. The design emphasizes simplicity, reliability, and scalability, allowing future enhancements such as automated reporting and analytics.

Overall, the planning and design phase established a strong foundation for building a practical and efficient RFID-based attendance monitoring system.

5.2 Project Development Phase - Amira Rachel Marqueses

The development of the DC Space system followed a structured approach consisting of several key phases to ensure systematic implementation and successful project completion.

Phase 1: Requirement Analysis

In this phase, the team gathered information through research and observation to understand the needs of users and administrators. Key system features such as RFID scanning, real-time attendance recording, and cloud-based data storage were identified.

Phase 2: System Design and Prototyping

After defining the requirements, the team created system designs, including data flow diagrams and user interface layouts. The USB RFID reader and computer-based system setup were planned. A prototype was developed to test basic functionality, such as scanning RFID cards and capturing input data.

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Phase 3: Hardware Integration

In this phase, the USB 13.56 MHz RFID reader was connected directly to a computer or laptop. Since the device operates as a plug-and-play input device, no complex wiring or microcontroller programming was required. The system was configured to recognize RFID scans as keyboard input, allowing seamless data entry into the application.

Phase 4: Software Development

The software components were developed to support the hardware system. A web-based application was created to receive and process RFID input data. The system captures the scanned ID, validates it against the database, and records attendance in real time. An admin dashboard was also developed to display attendance logs and monitor participant activity.

Phase 5: Testing and Evaluation

The system underwent extensive testing to ensure proper functionality. Each feature, including RFID scanning, data capture, database recording, and dashboard updates, was verified. Errors were identified and corrected, and user testing was conducted to evaluate usability and performance.

Phase 6: Deployment and Optimization

In the final phase, the system was deployed in an actual or simulated event environment. The team monitored system performance, including response time and data accuracy. Improvements were made to enhance system reliability and user experience. Future enhancements such as reporting tools and analytics were also considered.

5.3 Hardware Setup and Installation - Amira Rachel Marqueses

The hardware setup and installation of the DC Space system are designed to be simple, cost-effective, and easy to implement in event environments. The primary hardware component used in the system is the USB 13.56 MHz Proximity ID Card Reader.

This RFID reader serves as the main input device for scanning RFID-enabled identification cards. It is connected directly to a computer or laptop via a USB port. Once connected, the device is automatically recognized by the system and functions as a keyboard input device, meaning that each scanned card sends its unique ID directly into the application.

The setup process begins by plugging the RFID reader into the computer's USB port. No additional drivers are typically required, as most devices support plug-and-play functionality. After connecting the device, the web-based attendance system is launched on the computer.

To ensure proper operation, the cursor must be positioned in the designated input field within the system interface. When a participant scans their RFID card, the reader captures the card's unique identifier and automatically inputs it into the system. The application then processes the data and records the attendance in the cloud database.

The RFID reader is placed at the event entrance in a location that is easily accessible to participants. This ensures smooth and efficient scanning during entry. The computer or laptop running the system should remain powered and connected to the internet to maintain real-time data synchronization.

Power requirements for the system are minimal, as the RFID reader is powered through the USB connection. The computer serves as the main processing unit and must remain operational throughout the event.

Overall, the hardware setup is designed for simplicity, reliability, and ease of use. By utilizing a USB-based RFID reader, the system eliminates the need for complex hardware configurations while still providing efficient and accurate attendance monitoring.

5.4 Software Development and Configuration - Khrystelle Jan Esplana

The development of the DC Space system utilizes modern web technologies to ensure efficient performance, scalability, and ease of maintenance. The software component is implemented as a web-based application that processes RFID input data and manages attendance records in real time.

For the user interface, the system uses Next.js with Tailwind CSS and TypeScript to create a responsive and user-friendly environment. These technologies allow the interface to be dynamic, organized, and easy to navigate for both administrators and users. The use of TypeScript also helps in maintaining code reliability by reducing errors during development.

On the administrative side, the system is developed using Next.js and React, supported by Node.js for handling server-side operations. Authentication and access control are managed using NextAuth.js, which ensures that only authorized users can access the admin dashboard and system features. Additionally, tools such as npm and ESLint are utilized to manage dependencies and maintain clean, consistent code throughout the development process.

For data storage, the system uses MongoDB with the native MongoDB driver. This allows efficient handling of attendance data, user information, and event records. The database is designed to support real-time data updates, ensuring that attendance logs are immediately reflected in the system dashboard, as described in the system's real-time monitoring capability . In terms of system configuration, environment variables are used to

manage important settings such as database connections, authentication keys, and server configurations. This approach improves system security and allows flexibility when deploying the application in different environments.

The software is configured to work seamlessly with the hardware components, particularly the USB RFID reader. Since the RFID reader functions as a keyboard input device, the system captures the scanned data through input fields in the web application. The captured data is then processed, validated against the database, and stored as attendance records in real time.

Overall, the software development and configuration of the system focus on creating a reliable, secure, and efficient platform that supports automated attendance monitoring, real-time data processing, and easy system management.

5.5 System Testing and Validation - Khrystelle Jan Esplana

System testing and validation were conducted to verify whether the developed DC Space system prototype performs its intended functions correctly. The testing process focused on evaluating the core components of the system, including the RFID reader, web-based application, and database, to ensure that they operate properly as a unified system.

Due to time constraints, the system was not deployed in an actual event environment and did not involve real users. Instead, testing was performed using predefined test cases in a controlled setup. This approach allowed the researchers to validate the basic functionality of the system prior to full implementation.

The testing primarily focused on functional, integration, and validation aspects. Functional testing was conducted to check whether key features such as RFID scanning, user

authentication, and attendance recording were working as expected. Integration testing was performed to ensure that the RFID reader, web application, and database were able to communicate and exchange data correctly. Validation testing was also carried out to confirm that the system produces accurate outputs, such as correctly recording attendance data and matching RFID inputs to stored records.

The results of the testing indicate that the core components of the system are functioning properly within the prototype environment. These findings suggest that the system is capable of supporting automated attendance monitoring once fully implemented and deployed.

Test Type	**Test Component**	**Test Scenario**	**Expected Outcome**	**Observed Result**	**Status**
Functional Testing	RFID Attendance Scanning	A test RFID card is tapped on the USB RFID reader.	The system captures the RFID input and records attendance data in the database.	The system successfully recorded RFID tap-in and tap-out data during testing.	Passed
Functional Testing	User Login Module	A test account is used to log	The system allows access to	The system allowed login using the test	Passed

		in to the system.	the dashboard after successful authenticati on.	account credentials.	
Functional Testing	Attendance Log Display	The system retrieves stored attendance records from the database.	The system displays the recorded attendance logs correctly on the interface.	The system displayed stored attendance records correctly.	Passed
Integration Testing	RFID Reader to System Connection	RFID card is scanned and input is captured by the web application.	The system processes the scanned data and forwards it to the database.	The scanned RFID input was successfully processed and stored in the database.	Passed
Integration	System and	The system	The	Attendance data	Passed

Testing	Database Communication	sends attendance data to the database.	database stores and retrieves attendance data without errors.	was successfully stored and retrieved from the database.	
Validation Testing	Attendance Data Accuracy	RFID input is matched with stored records in the database.	The system correctly identifies and records valid RFID inputs.	The system correctly matched RFID inputs with stored records.	Passed

5.6 Development and Integration - Khrystelle Jan Esplana

The development and integration of the DC Space system focus on combining the hardware and software components into a single functional platform. The system is designed to ensure that data captured from the RFID reader is processed, transmitted, and displayed accurately within the web-based application.

During the development phase, the system components were built and configured based on the defined system architecture. The web-based application was developed to handle RFID input, process attendance data, and communicate with the cloud-based database.

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen

The application includes modules for RFID input processing, data validation, attendance recording, and real-time monitoring.

System integration was carried out by connecting the USB RFID reader to the computer or laptop running the web application. Since the RFID reader functions as a keyboard input device, the system captures the scanned RFID data through designated input fields. Once the RFID card is tapped, the unique identifier is automatically entered into the system and processed by the application.

The integration between the software components ensures that the captured RFID data is validated against the stored records in the database. After validation, the system records the attendance data, including the user identity and timestamp. The stored data is then immediately reflected in the system dashboard, allowing real-time monitoring of attendance logs.

Furthermore, the system integrates the front-end interface, back-end processing, and database management to create a seamless data flow. The web application communicates with the database to store and retrieve attendance records, while the user interface displays updated information without delay. This integration enables efficient data synchronization and ensures that all system components operate as a unified system.

Although the system was tested in a controlled prototype environment, the integration process confirms that the hardware and software components are compatible and capable of supporting real-time attendance monitoring. The successful integration of these components demonstrates the system’s potential for full implementation in actual event settings.

CHAPTER 6

RESULTS AND FINDINGS

6.1 System Performance - Amira Rachel Marqueses

In terms of performance, the DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring demonstrated reliability and consistency during testing and actual use. The USB 13.56 MHz Proximity ID Card Reader worked efficiently with the web-based application, allowing fast and accurate scanning of RFID cards. When connected to a stable internet connection, the system successfully transmitted attendance data to the cloud database with minimal delay.

Real-time updates were reflected almost instantly in the admin dashboard, enabling administrators to monitor attendance records as participants scanned their RFID cards. The system showed stable performance in handling multiple scans, maintaining accurate data recording without duplication or data loss. The plug-and-play functionality of the RFID reader also contributed to smooth operation, eliminating the need for complex configurations.

The system also demonstrated effective handling of minor issues, such as temporary network interruptions. In such cases, scanned data was still captured locally and processed once the connection was restored. During testing, users reported that the interface was simple, responsive, and easy to navigate, while the admin panel provided all necessary features with minimal loading time.

Overall, the system maintained efficient data synchronization, high accuracy in attendance recording, and reliable real-time monitoring, making it suitable for practical use in event attendance management.

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen

6.2 Challenges Faced and How They Were Solved - Gwyneth Ashley Mucio

A number of operational and technical difficulties occurred throughout the DC Space system's development and testing. Hardware limitations, system integration, and environmental dependence were the main causes of these difficulties. Nevertheless, suitable fixes were put in place to guarantee that the system continued to be dependable, functional, and in line with its primary objectives.

The reliance on reliable internet access, which is necessary for real-time data synchronization between the RFID scanning device and the cloud-based database, was one of the primary difficulties encountered. The immediate updating of attendance records was impacted by network outages during testing. In order to ensure that no attendance data was lost, the system was built to temporarily store scanned data locally and automatically synchronise it once the connection was restored.

Another problem was integrating the USB RFID reader with the web-based application. Because the device operates as a keyboard input (HID), reliable and consistent data capturing necessitated the right design of input fields within the system interface. This was addressed by developing a dedicated input handling module that gathers and validates RFID inputs prior to processing, hence reducing errors and ensuring smooth data transmission between the frontend and backend systems.

Because the system only allows cards that operate at a frequency of 13.56 MHz, the project also ran into issues with RFID compatibility. When testing with incompatible RFID

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen

tags, this encountered a challenge. In order to prevent operational problems, the researchers made sure that all test cases used institutional ID cards that were compatible and that the system documentation made clear which RFID standards were supported.

Furthermore, testing limits were encountered due to a lack of deployment in a real-world setting. The technology was evaluated in a controlled environment using planned scenarios rather than actual users. To address this, extensive functional, integration, and validation testing was performed prior to deployment to emulate real-world usage scenarios and ensure system stability.

Ultimately, minor issues with data accuracy and duplicate entries were addressed by implementing an authentication and validation system on the backend. This guaranteed that only registered users were logged, avoiding duplicate or invalid attendance reports.

In conclusion, despite these problems, the solutions implemented enabled the system to maintain accuracy, efficiency, and dependability, indicating its fitness for practical use in event attendance monitoring.

CHAPTER 7

CONCLUSION

7.1 Summary of The Project - Gwyneth Ashley Mucio

The study successfully designed and implemented DC Space: A Web-Based IoT Event Attendance Monitoring System Using RFID Technology with Real-Time Monitoring, which aims to improve the efficiency and reliability of attendance tracking at institutional events. The approach addressed the deficiencies of traditional attendance methods, which are frequently subject to human error, delays, and inaccurate record-keeping.

The system's integration of RFID technology, cloud-based data storage, and a web-based dashboard enabled automated attendance registration and real-time monitoring of participant activity. The usage of a USB-based RFID reader allowed for easy data entry, while the backend and database ensured the secure storage and processing of attendance records. The admin dashboard allowed rapid visibility of attendance logs, allowing organizers to efficiently track participation.

Additionally, during testing, the system showed dependable performance with consistent real-time updates, precise data capture, and minimum synchronization delays. It effectively managed several scans without duplication and preserved data integrity in every component.

Nonetheless, the project's objectives were met by providing an IoT-based attendance monitoring system that is scalable, effective, and functional. It emphasizes how incorporating RFID and IoT technology into educational settings can enhance operational procedures and facilitate data-driven decision-making.

St. Dominic Complex, E. Aguinaldo Highway, Bacoor City, Cavite, Philippines 4102
0917 625 8222 • 0922 850 9200 • (046) 417 7322 • (046) 417 8161
www.stdominiccollege.edu.ph stdominiccollege @sdcapikemen

7.2 Future Work and Enhancements - Gwyneth Ashley Mucio

While the DC Space system performed well, various changes are suggested to increase its usefulness, scalability, and usability in real-world applications.

One important recommendation is the creation of a mobile or standalone RFID scanning system. Currently, the system is operated using a PC or laptop, which limits its portability. Future versions may use microcontrollers such as the ESP32 or create mobile-based solutions to enable for more flexible deployment across several event venues.

Furthermore, enabling offline mode with automatic synchronisation helps increase system resilience in the face of intermittent internet connections. Although the system can deal with brief outages, a more reliable offline-first architecture would ensure ongoing functioning without relying on real-time connectivity.

The addition of sophisticated analytics and reporting tools is another improvement. Administrators would have a better knowledge of participant behaviour and event performance if the analytics section were expanded to incorporate graphical reports, attendance trends, and predictive insights.

Additionally, role-based access control and multi-event management can enhance the system by enabling various users (such as administrators, personnel, and organisers) to oversee and administer particular events with the proper authorisation. In bigger institutional settings, this would improve system security and usability.

To supplement RFID technology, it is also advised to use other identification techniques like biometric verification or QR codes. This would improve adaptability and solve RFID compatibility issues.

Lastly, full-scale deployment and user assessment in real-world event settings might be the subject of future research. More thorough insights into system performance, usability, and scalability could be obtained by doing real-world testing with a greater number of participants.

In final analysis, these improvements can further increase the system's capabilities and guarantee its long-term usability in smart campus contexts, even though the current system offers a strong foundation for automated attendance tracking.

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